

Aarhus University

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Acceptance Tests

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1. Summaries

The following section summarizes the results of the acceptance tests. The results are divided into two tables. One for the functional requirements, defined by the user stories and one for the non-functional requirements. Each table displays relevant information to the case under test.

1.1. Functional Test Summaries

This is the table for the functional requirements. Each requirement is listed with the ID of the user story, it's MoSCoW classification, a brief description of the requirement and a pass/fail grade.

User Story	MoSCoW	Requirement Description	Result
E1-US1: Player Movement	Must	- As a player - I want directional movement - So I can navigate the game environment horizontally	pass
E1-US2: Player (Cat) Jumping	Must	- As a player (cat) - I want to jump - So I can navigate the game environment vertically	pass
E1-US3: Player Collision	Must	- As a player - I want collision detection - So I cannot walk through obstacles	pass
E1-US4: Player Camera	Must	- As a player - I want a camera that follows my character - So I see the game world	pass
E2-US1: Realtime Visual Feedback	Must	- As a player - I want to be able to see the other player in real-time - So we can cooperatively solve the game challenges	pass
E2-US2: Pick Up Items	Must	- As a player (human) - I want to pick light-weight objects up - So I can solve puzzles	pass
E2-US3: Push/Pull Obstacles	Must	- As a player (human) - I want to push or pull heavy obstacles around - So I can solve puzzles	pass
E2-US4: Asymmetric Perspective	Must	- As a player - I want to observe the game environment in a way exclusive to my role - So that each player has a separate set of objectives	pass
E2-US5: In-Game Communication	Could	- As a player - I want to be able to give the other player a visual and audible clue - So I can communicate my intentions	pass

E2-US6: In-Game Voice Communication	Wont	• As a player • I want to be able to use in-game voice communication to interact with the other player • So we can coordinate our actions	fail
E3-US1: Level Timer	Could	• As a player • I want a timer to track my time used on a level • So that I can improve my best time	pass
E3-US2: Time Trial	Could	• As a player • I want levels to have a time goal • So the game provides a challenge	pass
E3-US3: Level Failure	Should	• As a player • I want to fail the level when I make critical mistakes • So the game provides challenge and consequence	pass
E3-US4: Level Failure Restart	Must	• As a player • I want the level to restart when I fail • So I can try again	pass
E3-US5: Player Life System	Should	• As a player • I want my character to have health • So I can be challenged and avoid taking damage	fail
E3-US6: Damage from Hazards	Could	• As a player • I want to take damage when I interact with dangerous objects • So I am challenged to avoid hazards	fail
E4-US1: Join Lobby	Must	• As a client • I want to be able to join a lobby • So I can play with a host	pass
E4-US2: Create Lobby	Must	• As a host • I want to be able to create an online lobby • So I can have a client join	pass
E4-US3: Role Selection	Could	• As a player • I want to choose my role • So I can play the chosen role	pass
E4-US4: Reconnect	Should	• As a host or client • I want to reconnect to a session after a disconnect • So I do not lose progress and won't have to start over	pass
E4-US5: Host Leave Level	Should	• As a host • I want to be returned to the lobby when I leave a level • So I can continue playing with the same client	pass

E4-US6: Client Leave Level	Should	- As a client - I want to return to the main menu when I leave a level - So I can start a new game	pass
E5-US1: Main Menu Navigation	Must	- As a player - I want a main menu with navigation options - So I can access game, settings and exit	pass
E5-US2: In-game Menu	Must	- As a player - I want an in-game menu - So I can return to the main menu	pass
E5-US3: Settings Menu	Could	- As a player - I want a settings menu - So I can adjust the game settings	pass
E5-US4: Guide Menu	Could	- As a player - I want a guide menu - So I can lookup the game interactions	pass
E6-US1: Level Progression	Should	- As a player - I want to complete levels with the help of another player - So I can progress in the game	pass
E6-US2: Auto Save	Could	- As a player - I want the game to automatically save my progression after a completed level - So I can continue my progression later	pass
E6-US3: Load Game Progress	Could	- As a player - I want to be able to load my saved game - So I can continue my progression	pass
E6-US4: Level Restart	Should	- As a player - I want to be able to reload the ongoing level - So I can try the same level if I get stuck	pass

1.2. Non-Functional Test Summaries

This is the table for the non-functional requirements. Each requirement is listed with a brief description of the requirement, it's MoSCoW classification and a pass/fail grade.

Furthermore, the tables are divided into each non-functional category of FURPS+.

Usability	MoSCoW	Result
A player is able to start a lobby in less than 3 clicks	Should	pass
The game is fully navigable using a keyboard and mouse	Must	pass
The game is fully navigable using a controller	Could	pass
The game provides a color-blind accessibility mode	Wont	fail
Game mechanics are introduced gradually	Should	fail

Reliability	MoSCoW	Result
Player positions are synchronized with no desync longer than 500 ms	Should	pass
The game automatically tries to reconnect players for 30 seconds, after a connection loss	Could	pass
The game will direct a player to the main menu, if a connection loss lasts more than 30 seconds	Should	fail

Performance	MoSCoW	Result
The game maintains at least 30 FPS on the target platform	Should	pass
Initial game load time is under 30 seconds	Could	pass
Network latency does not introduce more than 150 ms input delay	Should	pass
CPU usage should not exceed 80% on the target platform during peak gameplay	Could	pass
Scene transitions must complete within 5 seconds	Should	pass

Supportability	MoSCoW	Result
The game can receive updates post-release	Wont	fail
Players are able to report bugs	Wont	fail
The game monitors diagnostics during gameplay	Wont	fail

Security	MoSCoW	Result
The server validates all client inputs	Could	pass
User data complies with GDPR regulations	Wont	fail
The system must prevent basic client-side cheating by validating all game-play-critical actions server-side	Could	fail

2. Test Cases

This section presents the functional and non-functional acceptance tests. Each test is displayed in its own table, with information relevant to the case under test.

2.1. Functional Test Cases

The following section presents each functional test. The tests are divided by user story ID's and are being tested using the Gherkin test framework for software testing. The tests contain a brief Gherkin description of the expected outcome of the test, an observation of the actual test being performed and a pass/fail grade.

User Story	E1-US1: Player Movement
Description	• Given the player has spawned into the level • When the player presses the movement keys • Then the character should move in the chosen direction
Observation	The player was able to move freely around the environment using directional input.
Pass / Fail	pass

User Story	E1-US2: Player (Cat) Jumping
Description	• Given the player is controlling the cat • When the jump button is pressed • Then the cat should jump vertically
Observation	The cat successfully jumped and could reach elevated platforms.
Pass / Fail	pass

User Story	E1-US3: Player Collision
Description	• Given the player is near an obstacle • When the player walks into the obstacle • Then the player should not pass through it
Observation	Collision detection prevented the player from moving through walls and objects.
Pass / Fail	pass

User Story	E1-US4: Player Camera
Description	• Given the player is moving through the level • When the character changes position • Then the camera should follow the player • And the camera should collide with walls
Observation	The camera continuously followed the player and maintained visibility of the game world.
Pass / Fail	pass

User Story	E2-US1: Realtime Visual Feedback
Description	• Given two players are connected to the same session • When one player moves • Then the other player should see the movement in real-time
Observation	Player movement and actions were synchronized correctly between clients.
Pass / Fail	pass

User Story	E2-US2: Pick Up Items
Description	• Given the human player is near a light object • When the interact button is pressed • Then the object should be picked up • And the object should be set down when the interact button is pressed again
Observation	The player successfully picked up, carried and put down light-weight objects.
Pass / Fail	pass

User Story	E2-US3: Push/Pull Obstacles
Description	• Given the human player is near a heavy obstacle • When the interact button is pressed • And the player moves the character • Then the obstacle should move accordingly • And the obstacle should be detached when the interact button is pressed again
Observation	Heavy obstacles could be pushed and pulled, as well as let go to solve puzzles.
Pass / Fail	pass

User Story	E2-US4: Asymmetric Perspective
Description	• Given players are assigned different roles • When each player observes the environment • Then role-specific elements should be visible
Observation	Each role experienced unique gameplay information and interactions.
Pass / Fail	pass

User Story	E2-US5: In-Game Communication
Description	• Given two players are in the same session • When one player uses the communication feature • Then the other player should receive the clue
Observation	Audible communication cues were observed successfully.
Pass / Fail	pass

User Story	E2-US6: In-Game Voice Communication
Description	• Given two players are connected • When voice communication is activated • Then players should hear each other
Observation	Voice communication was not implemented in the prototype.
Pass / Fail	fail

User Story	E3-US1: Level Timer
Description	• Given a level has started • When the player is progressing through the level • Then a timer should track the elapsed time
Observation	The timer is tracking the time spent on a level, and is visible to the player on screen.
Pass / Fail	pass

User Story	E3-US2: Time Trial
Description	• Given the player starts a timed level • When the level is completed • Then the completion time should be compared to the goal time
Observation	The game tracked whether the player completed the level within the target time.
Pass / Fail	pass

User Story	E3-US3: Level Failure
Description	• Given the player makes a critical mistake • When the fail condition is triggered • Then the level should fail
Observation	Critical mistakes triggered the level failure state correctly.
Pass / Fail	pass

User Story	E3-US4: Level Failure Restart
Description	• Given the player has failed the level • When the restart condition is activated • Then the level should reload
Observation	Failed levels restarted successfully without requiring a full game restart.
Pass / Fail	pass

User Story	E3-US5: Player Life System
Description	• Given the player has health points • When the player takes damage • Then the health value should decrease
Observation	A life system was not implemented.
Pass / Fail	fail

User Story	E3-US6: Damage from Hazards
Description	• Given the player interacts with a hazard • When contact occurs • Then the player should take damage
Observation	No hazards were implemented, since the life system was not implemented.
Pass / Fail	fail

User Story	E4-US1: Join Lobby
Description	• Given a host has created a lobby • When the client enters the lobby code • Or when the client joins a public lobby • Then the client should join the session • And the client should be moved to the active game scene
Observation	Clients successfully joined hosted lobbies using the provided code.
Pass / Fail	pass

User Story	E4-US2: Create Lobby
Description	• Given the player is in the main menu • When the create lobby option is selected • Then a multiplayer lobby should be created • And a code for the lobby should be displayed on screen
Observation	Hosts could create online lobbies and wait for clients to join.
Pass / Fail	pass

User Story	E4-US3: Role Selection
Description	• Given two players are in a lobby • When the host selects a role • Then the chosen role should be assigned • And the client should be assigned the other role
Observation	Players were able to rotate between the available roles.
Pass / Fail	pass

User Story	E4-US4: Reconnect
Description	• Given a player disconnects during gameplay • When the player reconnects within the timeout period • Then the session should resume
Observation	Reconnection functionality partially restored disconnected players to the session.
Pass / Fail	pass

User Story	E4-US5: Host Leave Level
Description	• Given the host is in a level • When the host exits the level • Then the session should return to the lobby
Observation	The host returned to the lobby while maintaining the multiplayer session.
Pass / Fail	pass

User Story	E4-US6: Client Leave Level
Description	• Given the client is in a level • When the client exits the level • Then the client should return to the main menu
Observation	Clients could leave gameplay sessions and return to the main menu.
Pass / Fail	pass

2.2. Non-Functional Test Cases

This section presents the non-functional acceptance tests. Each test is comprised of the specific requirement under test, a description of the expected test outcome, the observation of the actual test being performed and a pass/fail grade.

Requirement	A player is able to start a lobby in less than 3 clicks
Description	• Given the player is in the main menu • When the player attempts to create or join a lobby • Then the action should be completed within three clicks
Observation	The lobby flow could be completed within the required number of interactions.
Pass / Fail	pass

Requirement	The game is fully navigable using a keyboard and mouse
Description	• Given the player is navigating the game menus • When keyboard and mouse inputs are used • Then all menus and gameplay features should remain accessible
Observation	All menus and gameplay interactions were accessible using keyboard and mouse controls.
Pass / Fail	pass

Requirement	The game is fully navigable using a controller
Description	• Given the player is using a controller • When navigating menus and gameplay • Then all essential interactions should function correctly
Observation	Controller support was partially implemented and functional for gameplay navigation.
Pass / Fail	pass

Requirement	The game provides a color-blind accessibility mode
Description	• Given the player opens the settings menu • When accessibility settings are inspected • Then a color-blind mode should be available
Observation	A dedicated color-blind accessibility mode was not implemented.
Pass / Fail	fail

Requirement	Game mechanics are introduced gradually
Description	• Given the player progresses through the game • When new mechanics are introduced • Then mechanics should appear incrementally over multiple levels
Observation	New mechanics were introduced progressively to avoid overwhelming the player. However, an enforced level order was not implemented.
Pass / Fail	fail

Requirement	Player positions are synchronized with no desync longer than 500 ms
Description	• Given two players are connected online • When players move simultaneously • Then player positions should remain synchronized within the allowed threshold
Observation	Multiplayer synchronization remained stable without noticeable long desynchronization periods.
Pass / Fail	pass

Requirement	The game automatically tries to reconnect players for 30 seconds, after a connection loss
Description	• Given a player loses connection • When the disconnect occurs • Then the game should attempt automatic reconnection for 30 seconds
Observation	The reconnect system attempted to restore the session after temporary disconnects.
Pass / Fail	pass

Requirement	The game will direct a player to the main menu, if a connection loss lasts more than 30 seconds
Description	• Given a player loses connection for longer than 30 seconds • When the reconnect timeout expires • Then the player should return to the main menu
Observation	Players were redirected to the main menu after prolonged connection loss.
Pass / Fail	fail

Requirement	The game maintains at least 30 FPS on the target platform
Description	• Given the game is running on the target platform • When gameplay is active • Then the framerate should remain above 30 FPS
Observation	Gameplay performance remained well above the minimum framerate target, at around 60 FPS during testing.
Pass / Fail	pass

Requirement	Initial game load time is under 30 seconds
Description	• Given the player launches the game • When the application loads • Then the game should start within 30 seconds
Observation	Initial loading times stayed below the required threshold on the target hardware.
Pass / Fail	pass

Requirement	Network latency does not introduce more than 150 ms input delay
Description	• Given players are connected over the network • When gameplay actions are performed • Then input delay should remain below 150 ms
Observation	Input responsiveness remained acceptable during multiplayer sessions.
Pass / Fail	pass

Requirement	CPU usage should not exceed 80% on the target platform during peak gameplay
Description	• Given the game is under peak gameplay load • When CPU usage is monitored • Then CPU utilization should remain below 80%
Observation	CPU usage remained within acceptable limits during gameplay tests.
Pass / Fail	pass

Requirement	Scene transitions must complete within 5 seconds
Description	• Given the player transitions between scenes • When a new scene is loaded • Then the transition should complete within 5 seconds
Observation	Scene loading times completed within the required duration.
Pass / Fail	pass

Requirement	The game can receive updates post-release
Description	• Given the game has been released • When updates are distributed • Then the game should support post-release updates
Observation	Post-release update functionality was not implemented in the prototype scope.
Pass / Fail	fail

Requirement	Players are able to report bugs
Description	• Given the player encounters a bug • When the player attempts to report it • Then a reporting feature should be available
Observation	An in-game bug reporting feature was not implemented.
Pass / Fail	fail

Requirement	The game monitors diagnostics during gameplay
Description	• Given gameplay is active • When diagnostics are monitored • Then gameplay statistics and diagnostics should be collected
Observation	Runtime diagnostic monitoring was outside the prototype scope.
Pass / Fail	fail

Requirement	The server validates all client inputs
Description	• Given a client sends gameplay input • When the server processes the request • Then the server should validate the action before applying it
Observation	Server-side validation was implemented for critical gameplay actions. However not for movement, to minimize the risk of rubberbanding.
Pass / Fail	pass

Requirement	User data complies with GDPR regulations
Description	• Given user data is stored or processed • When privacy compliance is evaluated • Then the system should comply with GDPR requirements
Observation	GDPR compliance was not implemented within the project scope.
Pass / Fail	fail

Requirement	The system must prevent basic client-side cheating by validating all gameplay-critical actions server-side
Description	• Given a client attempts to manipulate gameplay data • When the action is processed • Then invalid actions should be rejected by the server
Observation	Gameplay critical actions are performed on the server, but no actual validation is performed.
Pass / Fail	fail

3. Results

The acceptance tests showed that the Parallax prototype successfully fulfilled most of the core requirements defined at the start of the project. The main gameplay systems, including player movement, asymmetric perspectives, multiplayer synchronization and cooperative interactions, all worked as intended during testing.

The strongest results were seen in the multiplayer and gameplay systems. Players could create and join lobbies, move through levels together and interact with puzzles cooperatively. The asymmetric gameplay concept also worked successfully, as each player experienced different information and responsibilities within the same environment. This supported the project's main goal of encouraging communication and teamwork between players.

Performance-related requirements also produced positive results. The game maintained stable framerates during gameplay, scene transitions completed within the required time and multiplayer latency remained low enough to avoid major gameplay issues. The reconnect system was also able to reconnect disconnected players in most situations.

Several lower-priority systems were either incomplete or left outside the prototype scope. The health and hazard systems were not implemented, which caused related tests to fail. Likewise, features such as voice chat, accessibility options and GDPR compliance were not completed during development.

Overall, the tests confirmed that the prototype achieved its primary goal of creating a cooperative asymmetric multiplayer experience. Most failed tests were connected to optional or lower-priority features, while all of the important "Must" requirements passed successfully.

4. Discussion

The acceptance tests for Parallax were carried out on May 19th 2026, with all group members present. The tests were performed using the Gherkin method, which helped structure the tests around actual gameplay situations and expected outcomes.

The results showed that the project succeeded in delivering the core gameplay experience described in the original requirements. The asymmetric design, where players experience different versions of the same world and must communicate to solve puzzles, functioned as intended during testing.

Several of the project's core systems worked reliably during testing, including player movement, interaction systems, level progression and cooperative gameplay mechanics. The acceptance tests showed that players were able to complete levels and interact with the environment in the way originally intended. Performance during gameplay also remained stable, with acceptable framerate and loading times throughout the tests.

The tests also highlighted some of the project's limitations. Several gameplay systems, including health, hazards and advanced fail states, were not completed due to time constraints and changing priorities during development. Since the project was developed as a prototype rather than a full commercial game, the group focused mainly on the core gameplay loop and cooperative systems.

The MoSCoW prioritization method proved useful throughout development, as it helped the group focus on the most important features first. Even though some lower-priority requirements were unfinished, the acceptance tests showed that the main project goals were successfully achieved and that the prototype provides a solid foundation for future improvements.